

with palladium, are emerging as a cost-effective and reliable choice for advanced power electronics. For materials such as silicon carbide and gallium nitride, custom-coated or soft copper wires are being co-developed with customers to ensure better adhesion and long-term reliability.

Sustainability is also shaping this field, with 100% recycled gold bonding wire now available through advanced recycling processes that maintain high purity while supporting greener electronics.

How large is the potential for advanced materials in India?

India's semiconductor ecosystem is nascent but growing rapidly. Demand is expected to rise sharply, potentially surpassing China over time, although large-scale commercial production may begin around 2026-2028.

What is the global demand-supply situation for precious metals?

Post-Covid, electronics growth and US-China geopolitical tensions are prompting companies to diversify sourcing. Europe is focusing on recycling and urban mining. The US is offering incentives for local semiconductor production. China continues to dominate refining but faces environmental scrutiny. With strong global demand and limited supply, recycling is becoming increasingly important.

Who are the typical customers, and how do you approach the supply chain?

Our customers range from semiconductor fabs and OSATs to OEMs and recyclers. To serve them effectively, we maintain a resilient, regulation-compliant supply chain that ensures quality and traceability. Managing finite, high-cost metals while complying with regional regulations is essential to maintain consistent standards across markets.

What kind of partners or customers are sought?

Ideal partners possess expertise in

precious metals management, environmental compliance, recycling, and semiconductor technology. Collaboration spans OSAT companies, OEMs, and research institutions, especially where local regulations and waste management affect timelines.

Are advanced materials sourced from India or globally?

Most materials are currently procured from global manufacturing sites. We are exploring and assessing potential local partnerships for manufacturing or recycling as the next step towards further expansion.

Why do customers prefer Japanese suppliers?

Japanese suppliers are trusted for their rigorous quality systems, process discipline, and culture of continuous improvement. TANAKA's century-long expertise and end-to-end value chain—from refining to recycling—ensure quality, reliability, and traceability at every stage.

What innovations are being developed for advanced packaging?

Materials include gold, silver, copper (often palladium-coated), and aluminium wires for chip bonding and heat management. New trends focus on aluminium and copper wires for high-voltage and power electronics using substrates such as silicon carbide and gallium nitride. Solutions are co-developed with customers to ensure adhesion, reliability, and performance.

What challenges are faced in India?

Regulatory approvals for environmental compliance and waste management remain complex, making local partnerships essential for compliant manufacturing and smoother scale-up. A distributed global footprint helps mitigate risk during expansion.

Which policy changes could help?

Streamlined environmental and recycling permits, simplified import

regulations for precious metals, and incentives for recycling infrastructure would strengthen India's supply chain ecosystem.

What is the talent situation in India?

India's talent pool is strong and growing, with hires from top institutions and rising interest in semiconductor careers. The demand for skilled engineers in materials, electronics manufacturing, and recycling continues to expand, driven by both domestic and international opportunities.

Should Indian academia develop specialised curricula?

Yes. Advanced programmes in materials science, semiconductor manufacturing, recycling technology, and supply chain management will prepare students for electronics, sustainability, and global semiconductor roles.

How are recycling and sustainability handled?

Two main streams are used: production scrap and urban mining from end-of-life products. Metals are recovered using advanced extraction methods, analysed for purity, and refined. Gold bonding wires can now be produced entirely from recycled material, supporting sustainability and supply security.

How has the business mix evolved?

The focus has shifted from traditional assets to high-tech industrial solutions. Today, around 70% of revenue derives from electronics and industrial applications, including bonding wires, electrical contacts, and AI/data storage components. The goal is to grow sustainably across AI, automotive, and consumer devices. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/interview112025>

